Electrical Design Practical Guide

Lecture by Deputy Manager Atsuchi English Edition

15 October 2026

Introduction

This guide explains the material covered in the electrical design lecture, from the work of the Design Department and reading drawings to incoming power equipment, measurement, protection and control circuits, and handover to manufacturing. It is intended as a reference for people new to this work, to be read alongside the actual drawings.

Questions and answers from the lecture are also included. Terms and values that cannot be established from the lecture record are identified where they occur.

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Design Department organization and responsibilities

I belong to the Design Department. This is the department organization chart. The groups are organized under the department manager, and I belong to the electrical design group. The transcript refers to four groups; the supplied 2024 organization chart lists Electrical Design Sections 1, 2 and 3, the Special Assignments Section, and the Structural Design Section.

Electrical Design Section 1 prepares drawings while discussing various matters with customers. It is the busiest section in this group. Electrical Design Section 2 is a team of experienced staff who support Section 1. Its members have also previously worked in Section 1.

Electrical Design Section 3 handles low-voltage work. Sections 1 and 2 handle high-voltage work. The speaker names a person belonging to the Special Assignments Section, but that name cannot be determined from the transcript. The section mainly carries out development and also supports Section 1. It develops various programs and provides support so that the departments can work smoothly.

The Structural Design Section is the section we studied the other day. I belong to electrical design. Our daily work is busy because customer requests and matters within the factory often overlap. However, it is very rewarding. The name of the individual who prepared a drawing appears on it.

Additional points include Electrical Design Sections 1 and 2, Section 3, drawings such as single-line diagrams and outline drawings, document preparation, and communication with customers and other departments. Parts of the wording are unclear in the source. That concludes the explanation of the electrical design organization. Are there any questions? If not, let us move on.

Factory equipment and the flow of electricity

This is a drawing of the cubicle-type electrical installation at Kyoritsu Denki. As a basic convention, it is drawn so that electricity flows from top to bottom. This is the starting point of the supply. Up to this point the voltage is 6,600 V, on the high-voltage side. Below this point it is 210 V, on the low-voltage side.

These are the solar panels. Many solar panels are also installed on the roof of this building. The electricity generated by the solar panels is synchronized and connected with the supply received from the utility, and together they supply this facility.

This drawing represents everything with a single line. In other drawings, three lines are shown, and separate drawings show the control circuits that operate the equipment. The expression transcribed as “C flow” is unclear. The actual installation is over there. The products explained earlier are inside that enclosure.

The statement “This equipment is a generator” is immediately corrected to “It is not a generator.” If the supply to this installation fails, it can receive power from the generator. the generator is used as an emergency power supply during an outage, that there are also solar panels in front of the company, that electricity is not currently being sold, and that the company requires a large amount of power.

Design requests and original design drawings

In electrical design, we receive internal documents from Sales and proceed with the drawings. The exact formal name of the document is unclear in the transcript. We prepare the drawings based on the customer's manufacturing specification included in those documents.

We receive drawings similar to the one shown earlier. These drawings were prepared by a design office for the customer's building. We use them to prepare our first set of drawings. we check the components in the original design and any components that need to be added, and later make changes according to the customer's requirements.

This drawing shows the enclosure dimensions and overall size associated with the previous drawing. we check where the panels will be installed and their dimensions, and arrange them appropriately. The arrangement may start from the right or from the left, and in special cases the incoming panel may be in the middle. This table shows the breaker sizes. Some of the following numbers and interpretation are unclear in the source.

Drawing preparation revision history and specifications

I will now introduce the drawings we prepared based on the previous drawings. First is the cover sheet, which includes information about the customer. we submit the initial drawings to the customer and receive instructions on the changes required. This is the drawing revision history.

It can take up to two years to complete a set of drawings. The shorter period is transcribed only as “3…” and its unit cannot be determined. During that time, we hold various discussions with the customer and repeatedly prepare and submit drawings, revise them, and submit them again.

The question concerns whether the time or cost involved in drawing changes is charged to the customer through Sales. Parts of the question are unclear. The answer is that, as a rule, we do not charge for changing the drawings themselves. However, if additional items are required or the equipment becomes larger, we charge for that increase.

Next is the basic specification for the drawings. Our company manufactures customized equipment, so the products differ from one installation location to another. We therefore prepare the drawings to suit each location. In response to a question apparently asking whether this is only for design and assembly, the speaker explains that the specification is

also given to the customer to confirm, “We will proceed with this content.” The details of thequestion are unclear.

There are three sheets relating to these specifications. Next are the parts list and equipment list. They state items such as the equipment name, abbreviated symbol, rating, model, manufacturer and quantity. Some of the column names are unclear in the transcript.

Single line diagrams and high voltage incoming equipment

This is the skeleton diagram, or single-line diagram. We receive electricity from Kyushu Electric Power and show it flowing downward. First there is the disconnector. This is a switch. Further along is an arrester that diverts surge current to earth. Expressions such as “here” in the transcript refer to the positions being indicated on the drawing during the lecture.

This is the VCB, a vacuum circuit breaker. It also switches the circuit. If fault current flows while power is being supplied, the OCR, or overcurrent relay, detects it and switches the VCB off. This device converts a large current into a smaller current so that it can be read by the measuring instrument.

This is the automatic power factor controller. It receives current and voltage inputs and monitors the electrical condition. The transcript then says “turn VNC on,” but the device name and the relationship to the operation are unclear. Additional points include APFC and power factor values of “0.7” and “0.71,” but these cannot be established as setting values from the explanation.

The VT is a voltage transformer for instrumentation. It receives 6,600 V and converts it to a lower voltage for measurement. The transcript gives “100 V” and then “110 V” for the secondary voltage, so this value requires confirmation. This is the UVR, or undervoltage relay. This meter displays voltage, current, electrical energy and other quantities.

This device is the LBS, or load break switch. The transformer converts the high voltage of 6,600 V to the low voltage of 210 V. As in the previous example, a large current is converted to 5 A for measurement by the meter. The CT ratios and other numbers in the following interpretation are unclear in the source.

If too much current flows, this device operates and outputs an alarm. Its designation is transcribed unclearly as “30 re.” This number indicates the transformer capacity. A larger number allows a larger current. If this number changes, the fuse capacity, wire size, CT capacity and thermal relay setting also change.

Similar transformers are arranged in sequence: one, two, three and four.

Generator circuits and operation during a fire

This is the generator circuit. Normally, electricity is supplied from the transformer. During a power failure, the changeover device switches to the generator supply. The device name is unclear at this point in the transcript, but later it is called DT. Additional points include an emergency panel and loads such as ventilation, although parts cannot be deciphered.

If a fire occurs while power is being supplied by the generator and the sprinkler operates, this magnetic contactor, MC, is switched off. The load that operates the sprinkler is located here, and operation requires a large amount of power. Since this reduces the generator's available capacity, this switch turns off the circuit being indicated. The source does not explicitly identify the load being disconnected.

Because this drawing represents everything with a single line, we call it a skeleton diagram, or single-line diagram.

Breaker schedules operating flow diagrams and timing charts

Next is the breaker schedule. It shows items such as breaker frame size and trip current. The Japanese transcript gives “150 frame, 175 A trip,” while the following interpretation includes “250.” Since the values do not agree, they must be checked against the relevant drawing.

There are several sheets of these drawings. This is an operating flow diagram that simplifies the single-line diagram shown earlier. In automatic mode, if the utility supply fails, the UVR detects the failure and the switches are opened in sequence. When power returns, restoration is confirmed and the switches are closed again. This shows the broad sequence of operation.

In automatic mode, devices such as the VCB operate automatically according to the conditions. In manual mode, an operator goes to the installation to operate it. The details of the interpretation are unclear.

This timing chart expresses similar information against time. It shows when a circuit breaker is closed, opened and closed again. Additional points include reviewing and adjusting timing, such as the number of seconds before the generator starts after a power failure, and discussing it with the generator manufacturer and the customer. The details are unclear in the source.

Outline drawings layout drawings and installation dimensions

This drawing shows the dimensions and layout of the installation containing the products explained earlier. All dimensions are in mm and weights are in kg. Additional points include 900 kg, 800 kg and 1,200 kg, but the source does not identify the item associated with each value.

This is the front view, this is the rear view, and this is the side view. This is the plan view, or layout, looking down from above. Customers frequently refer to this drawing. Equipment such as the VCB, reactor and capacitor is arranged here. The equipment count and some names are unclear in the source.

The block skeleton diagram is drawn above the plan view. It simplifies the equipment positions, including the VCB and transformer, and shows the utility cables entering through the opening and following this route. This is the generator section discussed earlier.

These detailed dimensions relate to the anchor bolts. They show the holes used to bolt the equipment to the floor or foundation. This is the opening through which the customer brings in cables. customers frequently use this detailed drawing and can check views such as the side view. Parts are unclear in the source.

Individual panel details indications and operation

This drawing separates the installation into individual panels and shows each one in greater detail. There are outside and inside views. This is the fault annunciator. When a fault occurs, such as a power fuse fault, reactor abnormality or capacitor abnormality, the corresponding lamp illuminates and a signal is sent outside the panel.

This drawing is of the high-voltage incoming panel. It includes the VCB, disconnector, arrester, OCR, UVR and APFC, together with switches and indicators. Indications are provided when an undervoltage condition detects a power failure or when the overcurrent relay operates, for example. The speaker also explains an earth fault, meaning electricity flowing to ground. Some relationships between the indicators and equipment are unclear in the source.

This switch opens and closes the VCB, turning it off and on. This is the mode selector, which switches between manual and automatic operation. The interpretation also refers to meters, indicators, operating switches and manual/automatic selection.

These are the breakers. One of our company's strengths is a one-touch mounting arrangement that makes installation easy. The breaker is installed by inserting it. Additional points include four mounting methods, including a plug-in method and terms transcribed as “BB” and “F,” but the formal classifications and details are unclear.

Three line diagrams and measurement circuits

This is the three-line diagram. It shows the disconnector, VCB, OCR, APFC, UVR, multifunction meter and other equipment. The earlier drawings used a single line, whereas this drawing follows the actual wiring used when the equipment is manufactured in the factory.

Current and voltage lines are connected to this meter, and monitoring is carried out through an RS-485 communication line. voltage and current are obtained using a VT and CT and

used for calculations in the multifunction meter. Some of the calculated quantities areunclear in the transcript.

Everything previously shown with one line is represented with three lines here, including the LBS, transformer, CT and multifunction meter. The transcript includes “CT 302,” but it is unclear whether this is a designation or a numerical value.

This is a single-phase transformer for lighting, with a single-phase three-wire secondary. Two lines enter and three lines leave. The secondary voltage is 210 V between the two outer lines and 105 V between either outer line and the neutral. We prepare drawings like this for each transformer.

This is the generator circuit. As before, power is normally supplied through the transformer-side line. During a power failure, it is supplied from the generator. As explained earlier, this switch turns off the relevant circuit during a fire. Additional points include the normal-supply and generator-side terminals of the DT and the three-phase terminals of the MC. The details are unclear in the source.

Sequence control and protection circuits

We now move on to the sequence control drawings. I will explain what happens when the manual/automatic switch is set to automatic. First, there is an AC 100 V supply. When the switch is changed to automatic mode, the contact closes, current flows and the latching relay operates. The transcript also refers to positive and negative electricity, but the relationship to the AC supply is not clear.

We call this a keep relay, or latching relay. It turns on when set and off when reset. The drawing identifies the relay model and contacts, and indicates which drawings use the signals from the a-contacts and b-contacts. The model transcribed as “2D” requires confirmation.

Which relays are set and which are reset in manual and automatic modes. It also explains that references show where the contacts are used in subsequent drawings. Parts are unclear in the source.

This relay circuit provides indications and related actions when the undervoltage relay operates. Its designation is transcribed as “27” followed by an unclear character. This other circuit allows current to flow under the specified contact conditions in automatic mode and closes the VCB after the timer's set interval. The spoken description of the specific closing conditions is unclear.

We draw circuits of this kind. We connect circuits for the VCB, disconnector, lamps and other equipment, considering how to operate them to protect equipment during a fault and which switches should open to prevent the fault from spreading. We build the sequence circuits on that basis.

Even when the basic circuits are similar, we draw them according to the conditions and the customer's requested operation. This is followed by an explanation and correction concerning the tripping arrangement. Additional points include two methods, one obtaining DC from an AC supply and another using DC, but the formal names and their correspondence to the circuits are unclear.

Supply changeover control and external interfaces

We continue preparing drawings of this kind. This is the DC circuit, and this part is the DT/MC circuit. During a power failure, current flows through a b-contact to operate the changeover device. When power returns, one side is switched off and the other switched on to return to normal mode. The transcript also contains forms such as “DNC,” so some device designations are unclear.

The interpretation describes a signal for changeover to the generator first entering a timer and, after six seconds, operating through an “84 relay.” It then describes switching back to the utility supply when it is restored. The timing and relay number are as stated in the transcript and must be checked against the relevant drawing before being treated as actual settings.

Similar circuits follow, ending with the interface drawing. This drawing shows how the central monitoring system, or control center, is notified of which alarm has occurred when equipment develops a fault.

Ventilation calculations and handover to the factory

This is the calculation sheet. I will explain ventilation. Transformers become very hot, so we calculate what is required to keep the temperature inside the enclosure within the intended range. reviewing the number of ventilation openings and related provisions needed to maintain the temperature inside the panel. The source does not state a specific temperature range.

For the next drawing, the speaker says, “Let us leave this drawing out,” and skips its explanation. The final drawing is not submitted to the customer; it is shared only within the factory. It communicates points requiring attention. Since it is difficult to examine all the drawings in detail, these points are collected on one easy-to-read sheet.

Additional points include project-specific specifications such as incoming voltage and regional supply frequency. The source contains fragments corresponding to 6.6, 3.3, 50 and 60, but the units and the relationships between the values cannot be fully established.

In this way, the drawings are finalized and the equipment is manufactured in the factory. Are there any questions about what we have covered so far? Any questions are welcome.

Questions and closing

The participants refer to the drawing using expressions such as “this” and “here.” The interpretation contains words corresponding to “closed,” “open,” “normally closed” and “normally open,” but the full question and answer cannot be deciphered.

All right. If there are no questions, we will finish. Does anyone have any further questions? It appears that there are none. Part of the source, including an utterance transcribed as “central,” is unclear.

Was it difficult? Thank you for attending the lecture. Thank you very much. That was tiring. Thank you, everyone. You may leave now. The final utterance ends mid-sentence.

References and drawing examples

2024 Design Department organization chart (2 pages)
2024年 設計部新組織.pdf

Overall Kyoritsu factory drawing (1 pages)
共立工場全体.pdf

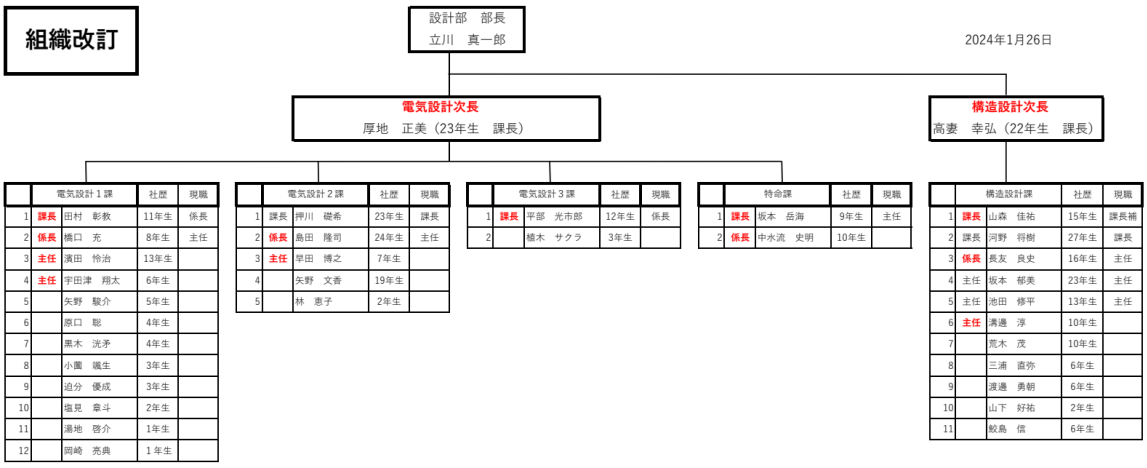
Goodman Takatsuki original design (6 pages)
②グッドマン高槻（原設計） 1 (1).pdf

Goodman Takatsuki electrical drawings (76 pages)
③グッドマン高槻（電気図面） 1 (1).PDF

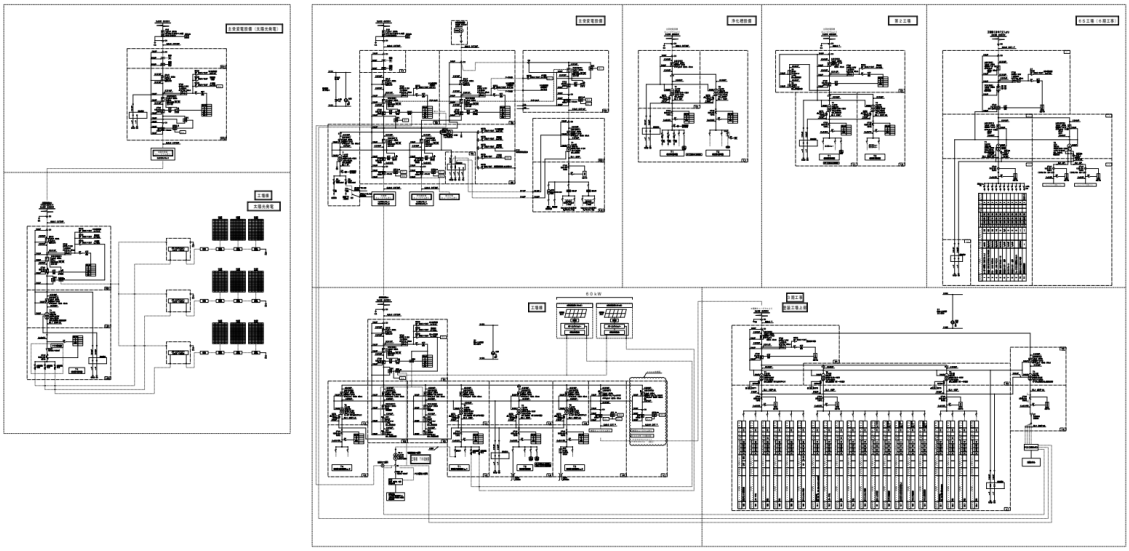
Goodman Takatsuki structural drawings (126 pages)
④グッドマン高槻（構造図面） 1 (1).pdf

Consult the supplied PDFs as the source drawings. The reduced illustrations below help identify the reference materials. Check dimensions and wiring details in the original PDFs.

2024 Design Department organization chart



Overall Kyoritsu factory drawing



Goodman Takatsuki original design

岡本

社内指示書

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受注番号 2021-04-C096
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受注分類 Z ﾊﾟﾅｿﾆｯｸ電工
建物区分
製図担当者A 850+350(ｽﾀﾝﾄﾞ工区)
製図担当者B
承認図納期日 2021/04/30

工作番号	製番品名	数量	製番単価	製番金額	盤名	希望納期日	売上No	売上金額
2021-003207	屋外QB 12面【TP, SC, SR支給品】	1式	¥10,900,000	¥10,900,000	QB	2C21/12/C1		
2021-003208	運送費	1式	¥600,000	¥600,000	送料	2C21/12/C1		
2021-003209	盤間接続費	1式	¥500,000	¥500,000	工事	2C21/12/C1		
			【合計】	¥12,000,000				¥0

できるだけ早く 松島様へメールお願いします。
岡本

副社長	相談役	常 務	営業部長	管理部長	設計部長	業 務	完了確認
		検 21.4.20 金丸	橋 21.4.19 口		21.4.23		

受注総額

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納入仕様書（図面）

作成　2021/04/26
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②
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件名　（仮称）グッドマン高槻新築工事　殿

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◎図中に記入している尺度は、図面の縮小・拡大に依り実尺と異なる場合がございますが、記入寸法を優先としますので御了承下さい。

図番番号	Z2104096/03207			
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販売：パナソニック株式会社エレクトリックワークス社
製造：株式会社　共立電機製作所

	グッドマン高槻新築工事			
①	正面⬇	左より1／12	H1	コンデンサ盤
②	正面⬇	左より2／12	H2	ＶＣＴ盤
③	正面⬇	左より3／12	H3	高圧受電盤
④	正面⬇	左より4／12	L1	低圧電灯配電盤Ｎｏ．1
⑤	正面⬇	左より5／12	L2	低圧電灯配電盤Ｎｏ．2
⑥	正面⬇	左より6／12	L3	低圧電灯配電盤Ｎｏ．3
⑦	正面⬇	左より7／12	L4	低圧動力配電盤Ｎｏ．1
⑧	正面⬇	左より8／12	L5	低圧動力配電盤Ｎｏ．2
⑨	正面⬇	左より9／12	L6	低圧動力配電盤Ｎｏ．3
⑩	正面⬇	左より10／12	L7	防災・保安動力配電盤
⑪	正面⬇	左より11／12	L8	保安電灯配電盤
⑫	正面⬇	左より12／12	H4	高圧分岐盤

ベース出荷時表記